

torch-spyre Ecosystem Health Report

Date: 2026-05-26 | Pod: rganti-spyre-main | PyTorch: 2.11.0+cpu

3/3 BUILD PASS	FAIL NIGHTLY CI	55 PRS MERGED (7D)	30 OPEN BUGS
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Build Status

Component	Commit	Status	Notes
deeptools	1969000189 — [DDC/ddl] use bitwise OR in identity so that it is type agnostic	PASS	cmake + make -j16 install (~13 min)
flex	c9e6f54 — perf(Issue #1077): remove using directive (#1075)	PASS	cmake + make -j16 install (~1 min)
torch-spyre	0155120 — Resolved missing attribute error in LX planning op tests (#2248)	PASS	uv sync editable install (~1m 44s)

CI Status

Workflow	Latest	Status	Trend
Spyre Hardware Tests	2026-05-26	PASS	Healthy
Linters	2026-05-26	PASS	Healthy
Upstream PyTorch Tests	2026-05-26	PASS	Healthy
Nightly Tests	2026-05-26	FAIL	Persistently failing (10+ days, 3 tests remaining)

Nightly Failure Triage

Failure 1 & 2: aten.cat with zero-sized tensors — REGRESSION Tests: test_cat_4d_dim2_zero, test_cat_4d_dim_m2_empty_first Error: InductorError: LoweringException: AssertionError — dim out of range for 1D empty tensor [0]

Root cause: PR #2259 ("Fix aten.cat lowering crash with zero-sized tensor inputs") merged May 22 but fix is ineffective in nightly environment. The lowering still receives a 1D [0] tensor and cat along dim=2/-2 triggers assertion.

Duration: 10+ consecutive days (since before May 17)

Assessment: **Regression — PR #2259 fix incomplete, needs lowering-level guard**

Failure 3: layernorm_2d_lx_planning_reduction — FLAKY

Test:

LxPlanningTwoOpReductionTest::test_layernorm_2d_lx_planning_reduction

Error: AssertionError: compiled spyre <-> cpu mismatch — 1/128 elements, abs diff 0.156 vs 0.1 tolerance

Root cause: Borderline fp16 precision in LX planning reduction mode. Rotating flaky — previously hit mish, now layernorm. Single-element outlier in fp16 multi-op compiled graph.

Duration: Intermittent (rotates between different ops in LxPlanningTwoOpReductionTest)

Assessment: **Flaky — tolerance too tight for fp16 + LX planning reduction**

Overall nightly: 3 failed, 2255 passed, 62 skipped, 1048 deselected, 323 xfailed, 2 xpassed. Pass rate: 99.87%.

Version Alignment

Check	Expected	Actual	Status
PyTorch	torch~=2.11.0 (pyproject.toml)	2.11.0+cpu	MATCH

Repo Drift (last 7 days)

Repo	Commits	API-touching	Risk
deeptools	82	10+ (DCC, BaseOpt, RcuOpt, HBM internals)	Low — compiler internals, no torch-spyre API surface changes
flex	8	6 (allocator refactor, MemoryType addition)	Medium — FlexAllocator::allocate signature change + MemoryType added
torch-spyre	43	—	—

Key flex changes: bbea5e3 adds MemoryType to FlexAllocator; 8f06481 removes default directive from FlexAllocator::allocate. Build passes — torch-spyre already adapted via PR #2190 (merged this week).

Blocking PRs

PR	Title	Why Blocking	Review Status
#2218	Upgrade to PT 2.12	Major dependency upgrade — blocks all PT 2.12 features	REVIEW REQUIRED

8 dependency PRs (dependabot) are stalled and represent accumulated CI/tooling debt.

Project Velocity

Metric	Value	Assessment
PRs merged (7 days)	55	Excellent — ~7.9 PRs/day
PRs in active review	18	Healthy pipeline
PRs approved (ready to merge)	3	#2180, #2269, #2274
Stalled PRs (>7 days)	36	Needs triage — many community contributions
Draft PRs	11	Normal

Open Bugs

Metric	Value
Total open bugs	30
New (last 24h)	0
Filed (last 7 days)	6 (propagate_hints x2, logical_not, Mistral-Small, MHA, Mistral unexpected arg)
Closed (last 7 days)	5 (torch.ne, dtype cast, expand, getitem, BMM dim labels)
Net change	+1 (roughly balanced)
Category breakdown	22 model-enablement, 8 inductor, 4 backend-compiler

Smoke Test

Test	Status	Details
Import & device detection	PASS	Spyre available, 1 device, torch 2.11.0+cpu
Eager execution (fp16 add)	PASS	4x4 tensor add on spyre device
Compiled execution (torch.compile)	PASS	128x128 fp16 add via Inductor/Spyre backend
Model E2E (Qwen3 0.6B)	PASS	5 tokens, load 2.9s, generate 73.9s — output: "Paris. The capital of"

Action Items

- 1 **[P1] Fix nightly cat regression:** PR #2259 is ineffective — investigate why zero-sized tensor filtering doesn't prevent the lowering assertion. Likely needs a guard at the Inductor lowering level, not just decomposition.
- 2 **[P2] Fix nightly layernorm tolerance:** Relax atol for LxPlanningTwoOpReductionTest (fp16 multi-op graphs) from 0.1 to 0.2, or pin random seed.
- 3 **[P2] Merge 3 approved PRs:** #2180 (eq scalar), #2269 (test docs), #2274 (CI fix) — all approved, ready to land.
- 4 **[P3] Prevent transformers downgrade:** Add post-build step to re-install hf_adapters deps after uv sync (transformers>=5.0 gets clobbered by lockfile).
- 5 **[P3] Review PT 2.12 upgrade (#2218):** 5 days old, high-impact — needs thorough review before merge.
- 6 **[P3] Triage stalled PRs:** 36 PRs have no activity >7 days — close abandoned ones, request rebases on viable ones.